

# Quadratic functions - Set 3 - Solutions

Specification MM1.3, M4.5, M4.11-M4.14.

## Answer key

| Question | Answer | Topic               |
|----------|--------|---------------------|
| 1        | D      | Quadratic functions |
| 2        | H      | Quadratic functions |

## Worked solutions

### Question 1 QUA-22 · Quadratic functions · spec MM1.3

The equation  $x^2 + ax + b = 0$  has roots 3 and  $-5$ . The equation  $x^2 + bx + a = 0$  has roots  $p$  and  $q$ . Find  $p^2 + q^2$ .

**Answer: D**

Sum of roots  $3 + (-5) = -a$ , so  $a = 2$ ; product  $-15 = b$ . Then  $x^2 - 15x + 2 = 0$  has  $p + q = 15$  and  $pq = 2$ , so  $p^2 + q^2 = 15^2 - 2(2) = 221$ .

#### Full method - how and why

Two linked quadratics: the roots of the first give the coefficients of the second. Sum-and-product relations do all the work.

- 1 For  $x^2 + ax + b = 0$  with roots 3 and  $-5$ :  $\text{sum} = 3 + (-5) = -2 = -a$ , so  $a = 2$ .  
*why:* Sum of roots  $= -b/a$ ; with a monic quadratic that is just  $-a$  - mind the double negative.
- 2 Product  $= 3 \times (-5) = -15 = b$ .  
*why:* Product of roots  $= c/a = b$  here.
- 3 The second equation is  $x^2 - 15x + 2 = 0$ , so  $p + q = 15$  and  $pq = 2$ .  
*why:* Substitute the values found, being careful that the roles of  $a$  and  $b$  swap between the two equations.
- 4 Use  $p^2 + q^2 = (p + q)^2 - 2pq = 225 - 4 = 221$ .  
*why:* The same identity as in QUA-04.

#### Where it goes wrong

- Getting  $a = -2$  by forgetting the minus in 'sum  $= -a$ ', which turns the second equation into  $x^2 - 15x - 2$  and gives 229.
- Computing  $(p + q)^2$  and forgetting to subtract  $2pq$ , giving 225.

**Takeaway** Read coefficients from roots with  $\text{sum} = -b/a$  and  $\text{product} = c/a$ , and keep track of which equation you are in.

## Question 2 X2-01 · Quadratic functions · spec MM1.3

$p$  and  $q$  are real numbers. Which of the following statements is/are true?

- I If  $p^2 < 4q$  then  $x^2 + px + q > 0$  for all real  $x$ .
- II If  $x^2 + px + q > 0$  for all real  $x$  then  $q > 0$ .
- III If  $q < 0$  then  $x^2 + px + q = 0$  has two real roots of opposite sign.

**Answer: H**

I: negative discriminant and positive leading coefficient. II: put  $x = 0$ . III: the product of the roots is  $q < 0$  (and the discriminant  $p^2 - 4q > 0$ ), so the roots are real with opposite signs.

### Full method - how and why

Three claims about a quadratic. Each is settled by the discriminant, by a well-chosen substitution, or by the root-product relationship.

- 1 I:  $p^2 < 4q$  means the discriminant  $p^2 - 4q < 0$ , so  $x^2 + px + q$  has no real roots; with a positive  $x^2$  coefficient the graph lies entirely above the axis. **True.**  
why: No roots plus an upward parabola means always positive - the graph never reaches the axis.
- 2 II: if  $x^2 + px + q > 0$  for **all**  $x$ , then in particular at  $x = 0$ :  $q > 0$ . **True.**  
why: Substituting a convenient value is the quickest way to extract information from a universal statement.
- 3 III: the product of the roots is  $q$ . If  $q < 0$  then  $p^2 - 4q > p^2 \geq 0$ , so the roots are real and distinct, and their product being negative makes them opposite in sign. **True.**  
why: For  $x^2 + px + q$  the sum of roots is  $-p$  and the product is  $q$ .
- 4 So all three.  
why: Match to the list.

#### Where it goes wrong

- Reading I's condition as  $p^2 > 4q$  - that would give two real roots and the opposite conclusion.
- Forgetting to check in III that the roots are **real**; a negative product alone would be meaningless for complex roots.
- Trying to prove II with a general argument when  $x = 0$  settles it in one line.

**Takeaway** Discriminant controls the number of roots;  $x = 0$  extracts the constant term; sum and product of roots are  $-b/a$  and  $c/a$ .